

Questions with Answer Keys

MathonGo

Q1: 16 March (Shift 1) - Single Correct

Given below are two statements: one is labelled as Assertion A and the other is labelled as

Reason R :

Assertion **A** : The $\text{H} - \text{O} - \text{H}$ bond angle in water molecule is 104.5°

Reason **R** : The lone pair — lone pair repulsion of electrons is higher than the bond pair - bond pair repulsion.

- (1) A is false but R is true
- (2) Both A and R are true, but R is not the correct correct explanation of A
- (3) A is true but R is false
- (4) Both A and R are true, and R is the correct explanation of A

Q2: 17 March (Shift 1) - Single Correct

A central atom in a molecule has two lone pairs of electrons and forms three single bonds. The shape of this molecule is:

- (1) see-saw
- (2) planar triangular
- (3) T-shaped
- (4) trigonal pyramidal

Q3: 17 March (Shift 2) - Single Correct

Amongst the following, the linear species is:

- (1) NO_2
- (2) Cl_2O
- (3) O_3
- (4) N_3^-

Q4: 18 March (Shift 1) - Numerical

Questions with Answer Keys

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AX is a covalent diatomic molecule where A

and X are second row elements of periodic table. Based on Molecular orbital theory, the bond order of AX is 25. The total number of electrons in AX is _____ (Round off to the Nearest Integer).

Q5: 18 March (Shift 2) - Single Correct

The oxide that shows magnetic property is :

(1) SiO_2

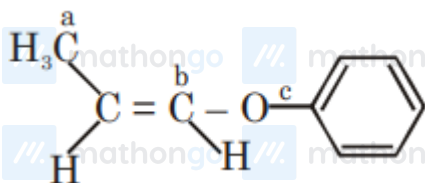
(2) Mn_3O_4

(3) Na_2O

(4) MgO

Q6: 18 March (Shift 2) - Single Correct

In the following molecules,



Hybridisation of carbon a, b and c respectively are :

(1) sp^3 , sp , sp

(2) sp^3 , sp^2 , sp

(3) sp^3 , sp^2 , sp^2

(4) sp^3 , sp , sp^2

Q7: 18 March (Shift 2) - Numerical

Questions with Answer Keys

MathonGo

The number of species below that have two

lone pairs of electrons in their central atom is _____

(Round off to the Nearest integer)

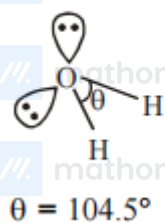
SF_4 , BF_4^- , ClF_3 , AsF_3 , PCl_5 , BrF_5 , XeF_4 , SF_6

Answer Key**Q1 (4)****Q2 (3)****Q3 (4)****Q4 (15)****Q5 (2)****Q6 (3)****Q7 (2)**

Hints and Solutions

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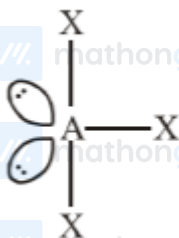
Q1 (4)



the hybridisation of oxygen in water molecule is sp^3 .

So electron geometry of water molecule is tetrahedral and the bond angle should be $109^\circ 28''$ but as we know that lone pair-lone pair repulsion of electrons is higher than the bond pair-bond pair repulsion because lone pair is occupied more space around central atom than that of bond pair.

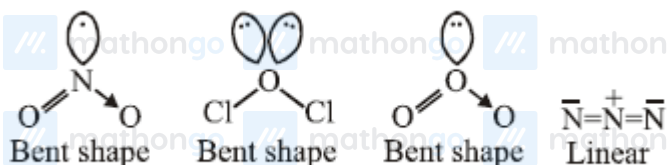
Q2 (3)



sp^3d hybridised

T-shaped

Q3 (4)



Q4 (15)

AX is a covalent diatomic molecule. The molecule is NO. Total no. of electrons is 15.

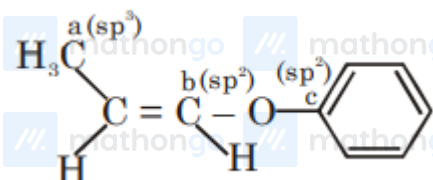
Hints and Solutions

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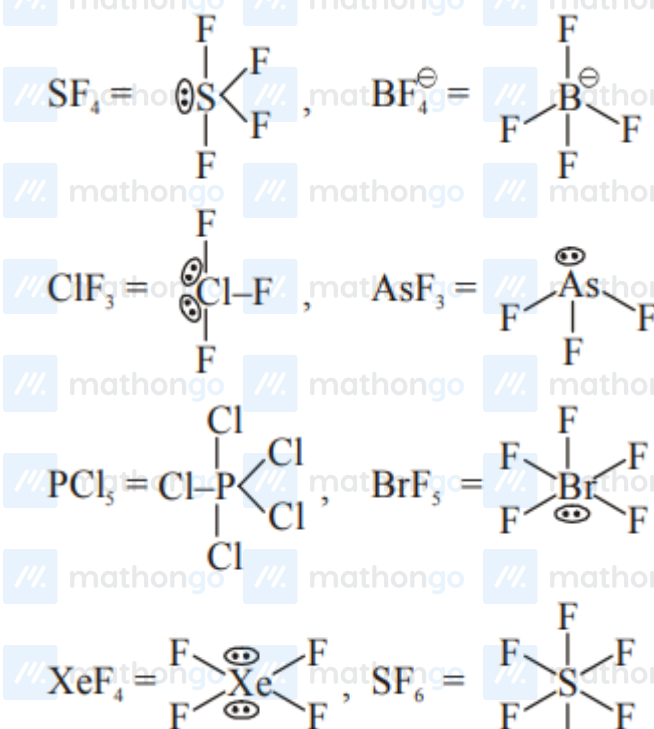
Q5 (2)

 Mn_3O_4 shows magnetic properties

Q6 (3)



Q7 (2)

Two l.p. on central atom is = $\text{ClF}_3, \text{XeF}_4$